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College of Engineering and Computer Science

AI in Action

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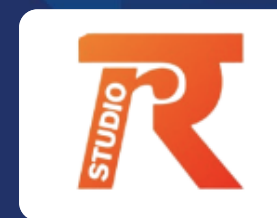
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AI Models to AI Products:

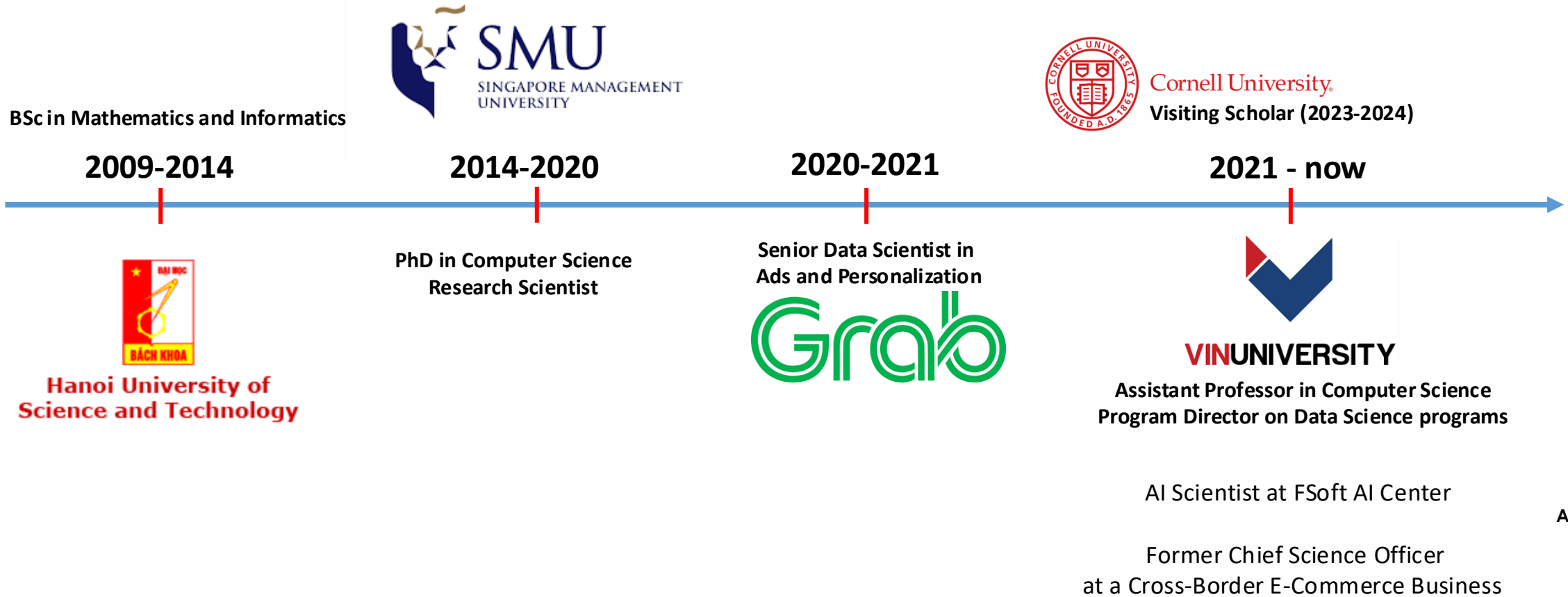
A path towards building models with real-world impact

Dr. Le Duy Dung

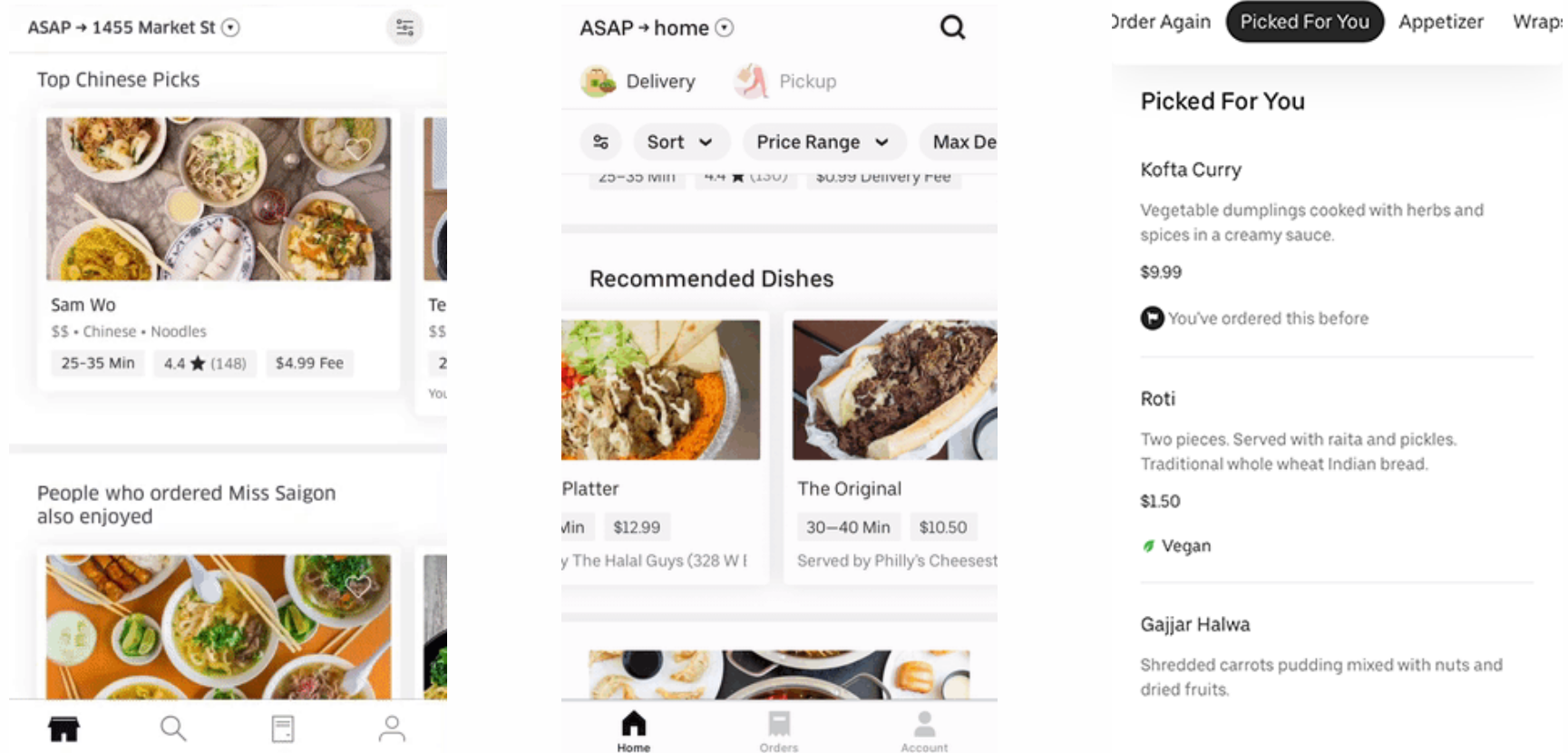
Assistant Professor in Computer Science,
Program Director of B.Sc. in Data Science,
Associate Program Director of B.Sc. in Computer Science,
College of Engineering and Computer Science, VinUniversity



About me!



An AI Developer Food Discovery Engine at Grab



Grab Food UI surfaces a Wealth of Options for hungry users informed by past orders and previously specified user preferences.

About R2Studio!



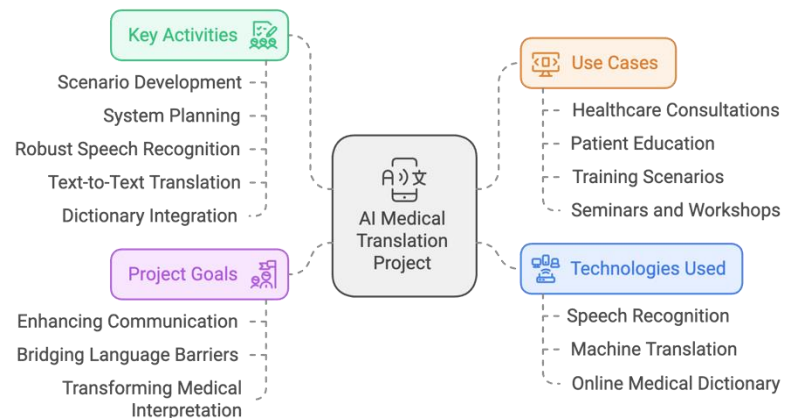
Research focus

AI-powered Search and Recommendation

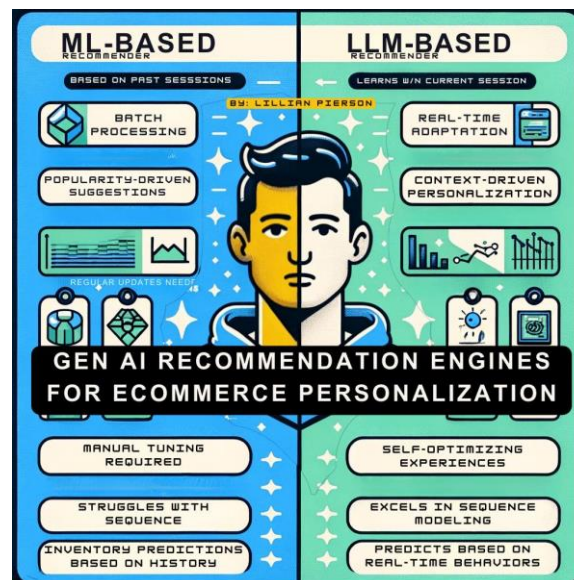
At R2Studio, we focus on pushing the advance of AI and machine learning techniques in the domain of search (retrieval) and recommendation.

We publish at reputable venues:

- Machine learning: ICML, NeurIPS, Machine Learning
- Artificial Intelligence: AAAI, IJCAI, JAIR
- Data Mining: WWW, SIGIR, WSDM
- NLP: ACL, EMNLP, COLING



Robust Vietnamese-English Clinical and Educational Medical Translation

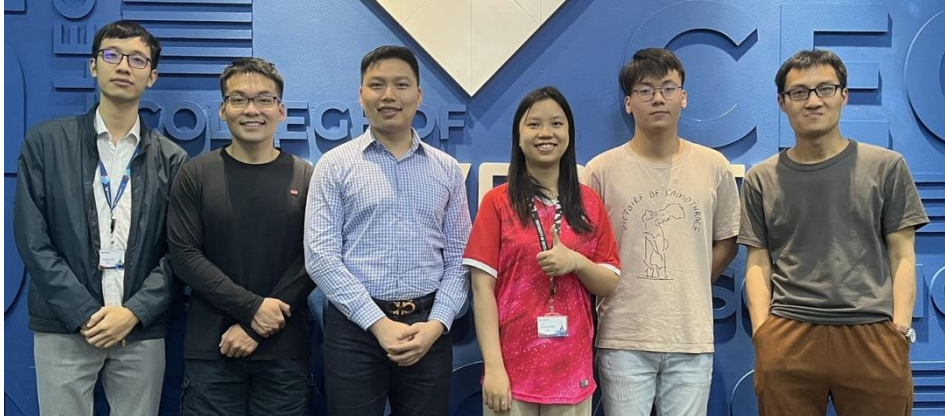


Multimodal Recommendation with Generative AI (LLMs, Diffusion Models, etc.)



AI and Data Science for Personalized Gifting

About V-IndoorCare!



VinUni-IndoorCare Group

(<https://vinuni-indoorcare.github.io/>)

VinUni-Illinois Smart Health Center

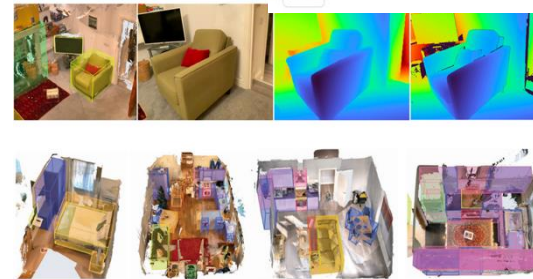
- Growing importance of indoor air quality in urban living
- Role of technology in **Monitoring** and **Enhancing** Indoor Air Quality
- The **transformative impact of AI** on traditional air quality management methods



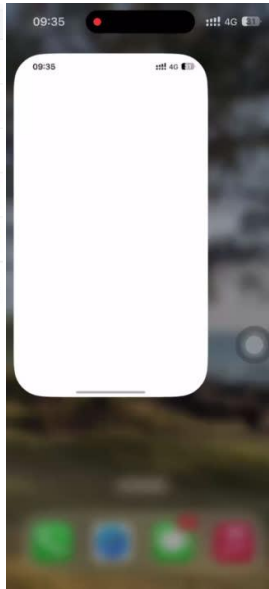
Air Sensors

Device	Connection	Environment	Fan speed	Particle count (L)	AQI (US)	PM2.5 (µg/m³)	PM10 (µg/m³)	PM1 (µg/m³)	CO2 (ppm)
AirVisual Pro - Vin04 6NMYR3	Wi-Fi	Indoor	--	--	56	12	--	--	499
AirVisual Pro - Vin03 4CJRGH	Wi-Fi	Indoor	--	--	58	13	--	--	487
AirVisual Outdoor - VinOutdoor 6T5UT2ZQK9	4G	Outdoor	--	--	--	--	--	--	--
AirVisual Pro - Vin02 5YUUL78	Wi-Fi	Indoor	--	--	55	11	--	--	487
AirVisual Pro - Vin01 8RU4JXR	Wi-Fi	Indoor	--	--	68	18	--	--	495

1-5 of 5 items 1 Jump to page:



RGB-D data



Lidar Sensor



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Section 1: Going beyond Model.fit() to also optimize for Market.fit()

Dr. Le Duy Dung

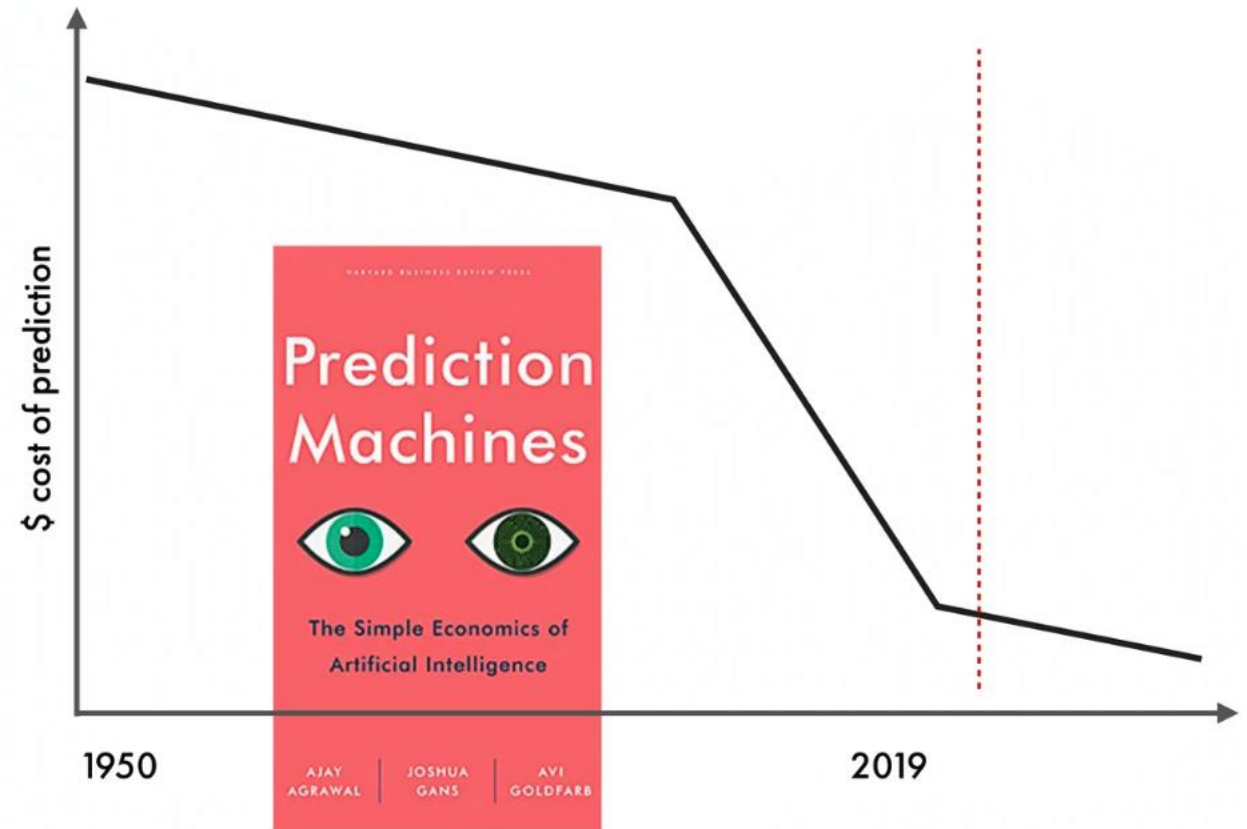
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An overview of AI systems

The simple economics of AI/ML

AI can reduce the cost of prediction –
this means that more businesses will aim
to turn their existing problems into prediction
problems



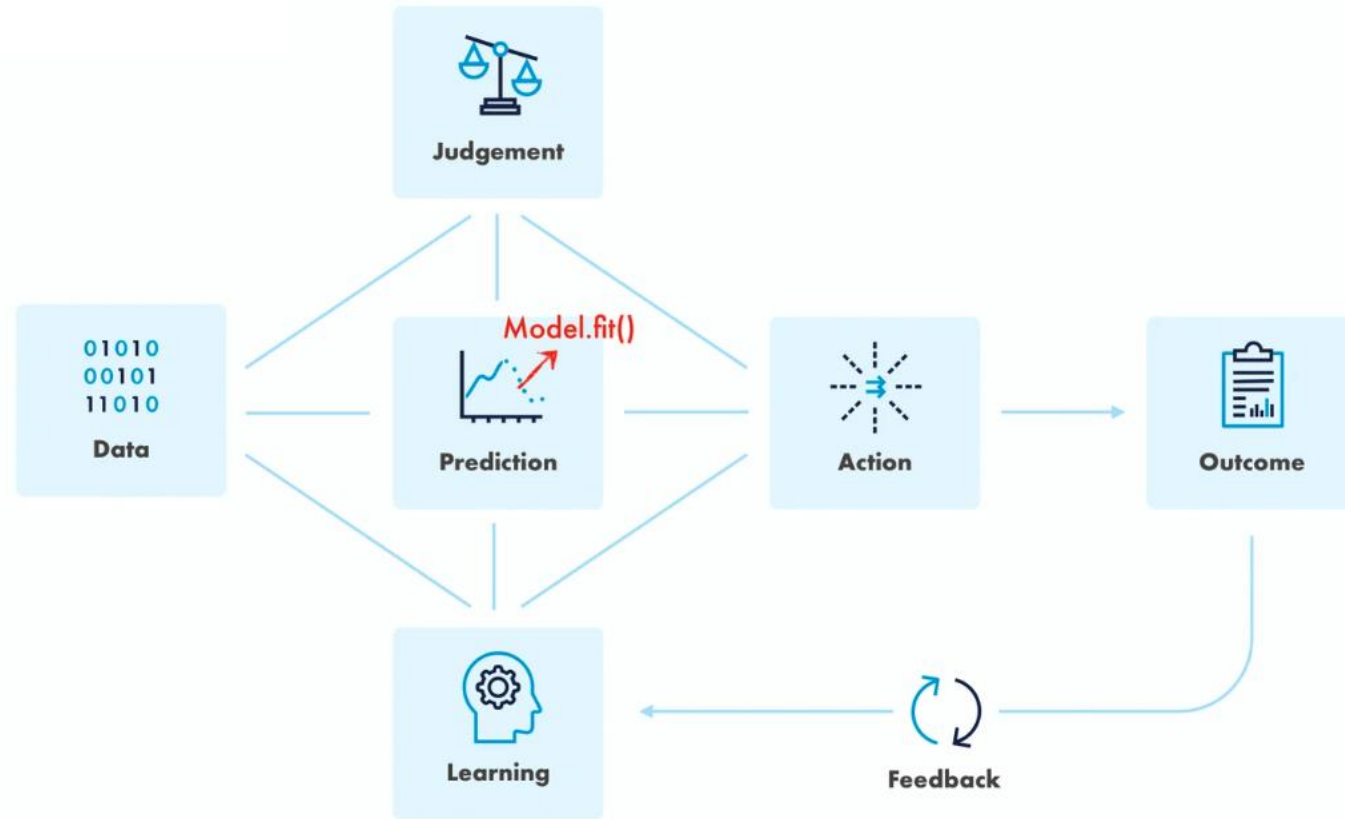
Reflection!

Which problems in your industry has been turned into **prediction** problems?

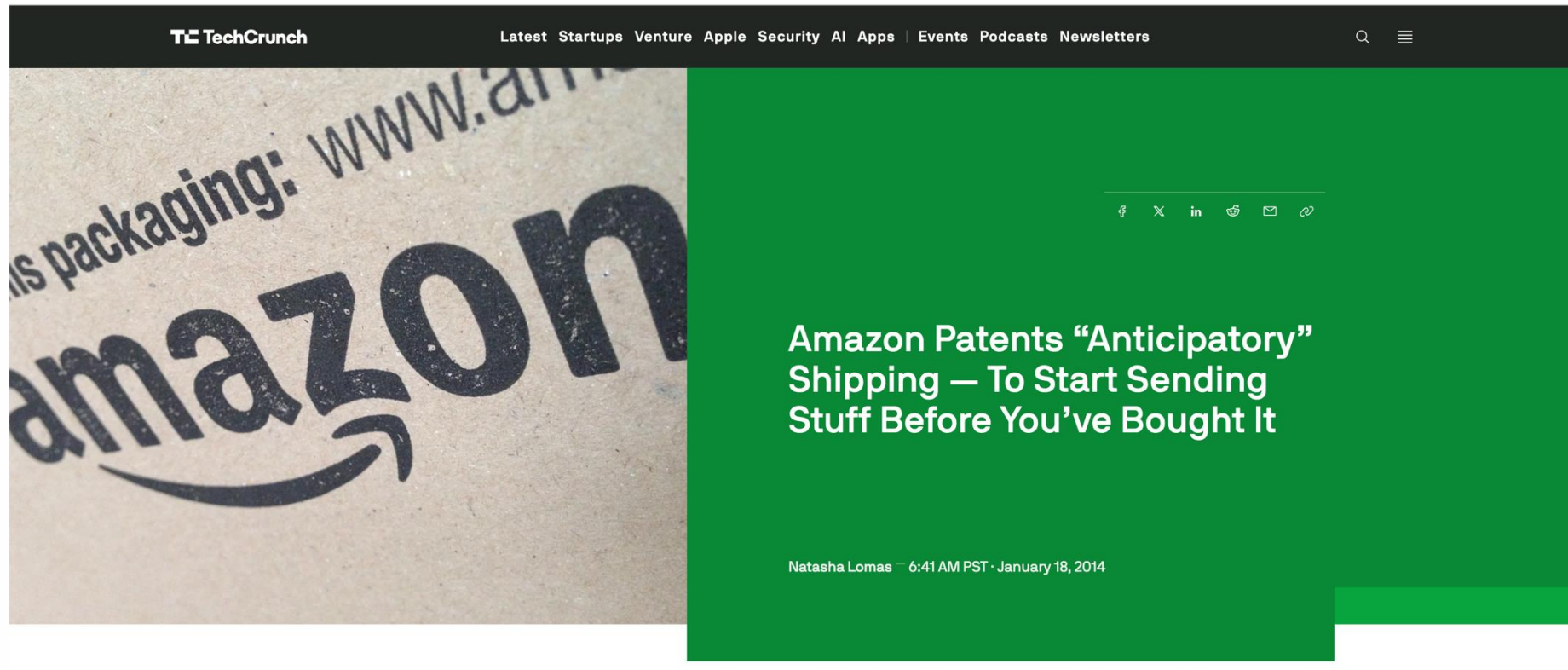
AI is a component in a decision-making system

The new economics of prediction will have an impact on the economics of other components in the system.

For instance, cheaper prediction is likely to lead to more expensive judgment, or data. Therefore, the impact of AI should not be considered in isolation, and rather as part of a system.

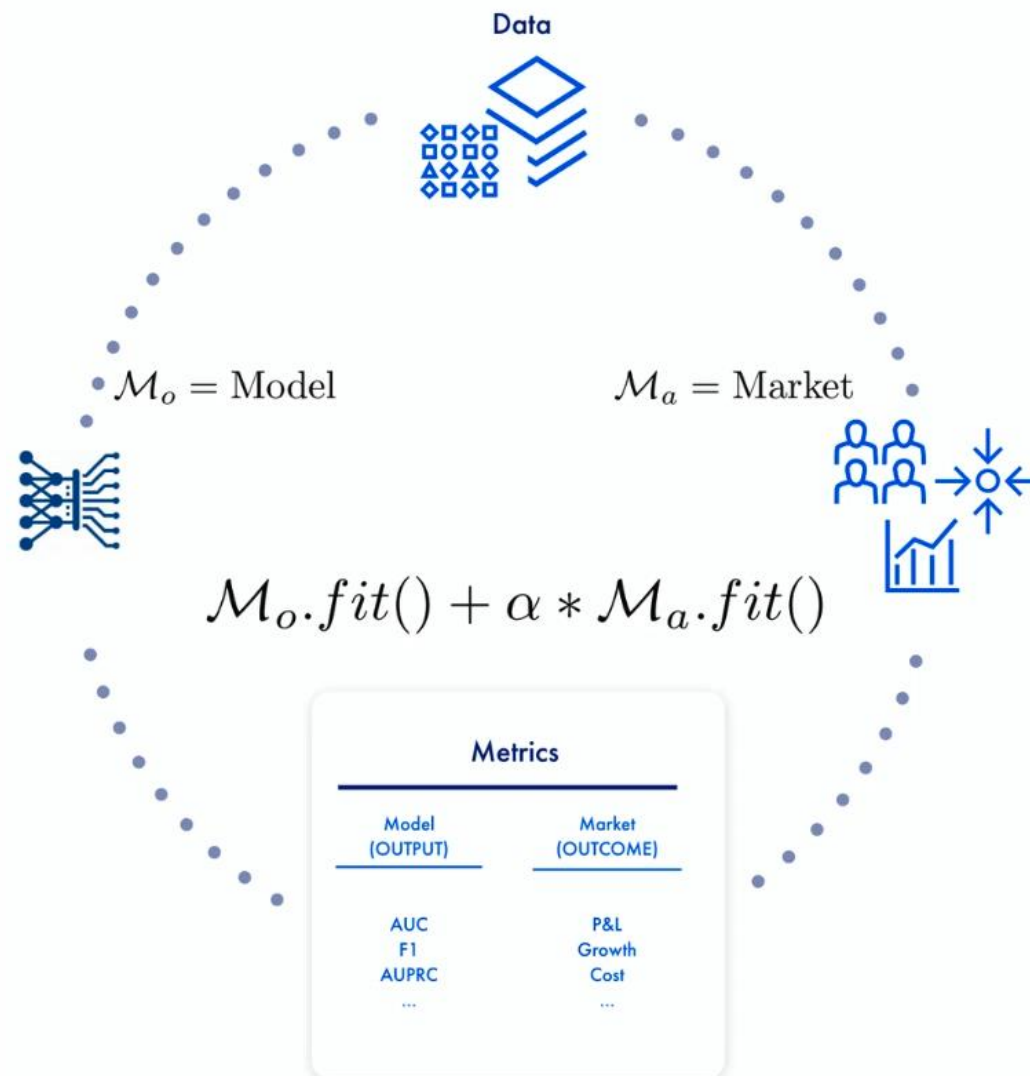


An example of AI-first transformation in Retail



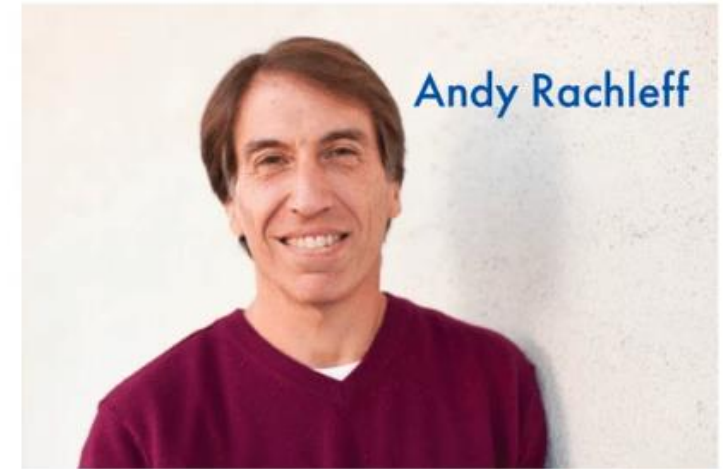
Switching to anticipatory shipping, for instance, leads to a different company/product; not reliant on the same approach to customer acquisition and user experience, and a much stronger reliance on ML's ability to correctly anticipate.

This session



A product is a specific offering intended to meet a set of customer needs

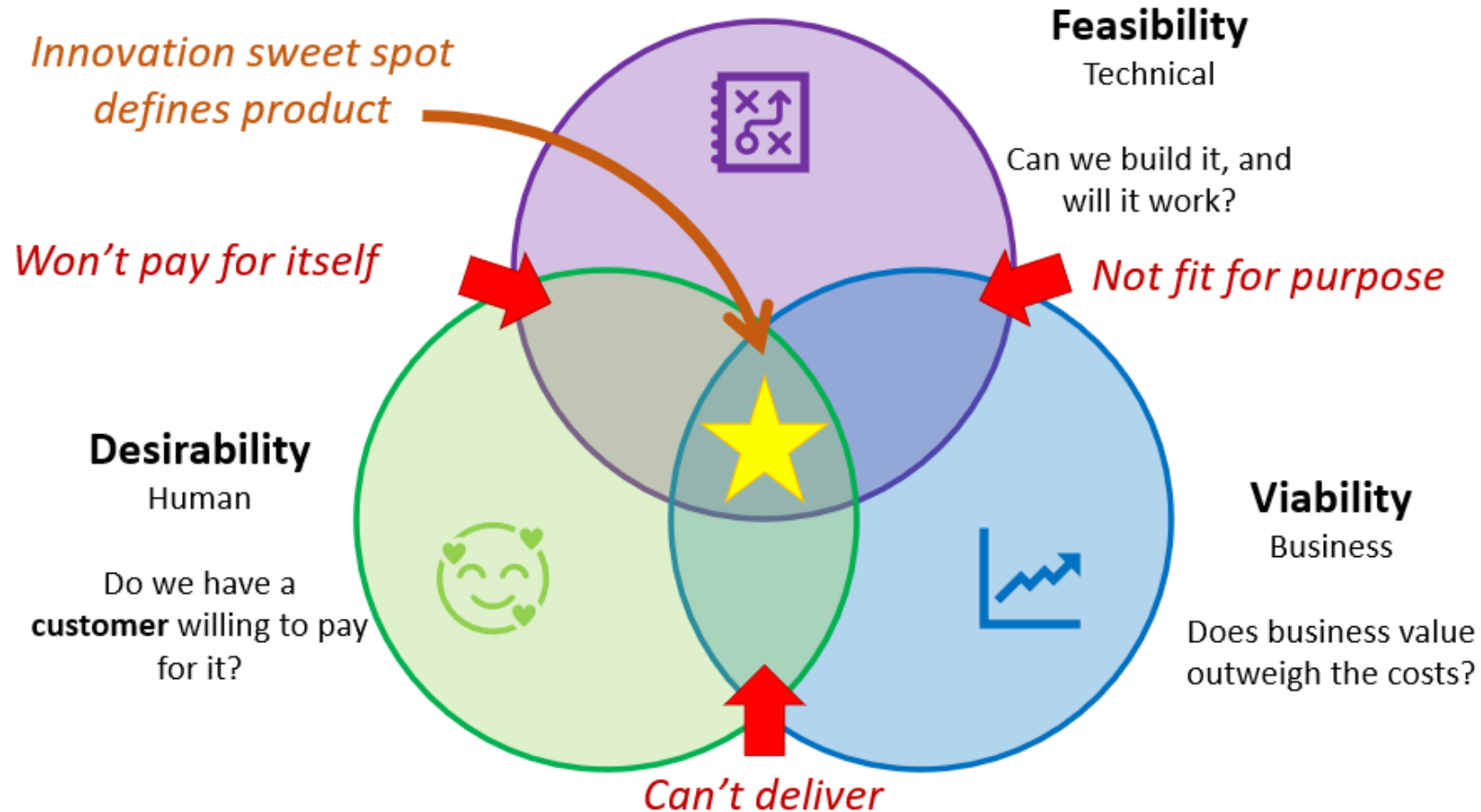
The goal is to not leave things to chance and rather devise a systematic approach to discover, design, and deliver



When a great product meets a lousy market, market wins.
When a lousy product meets a great market, market wins.
When a great product meets a great market, something special happens.

$$P(fit|\mathcal{M}_o, \mathcal{M}_a) \approx P(fit|\mathcal{M}_a)$$

A product needs to optimize for ...



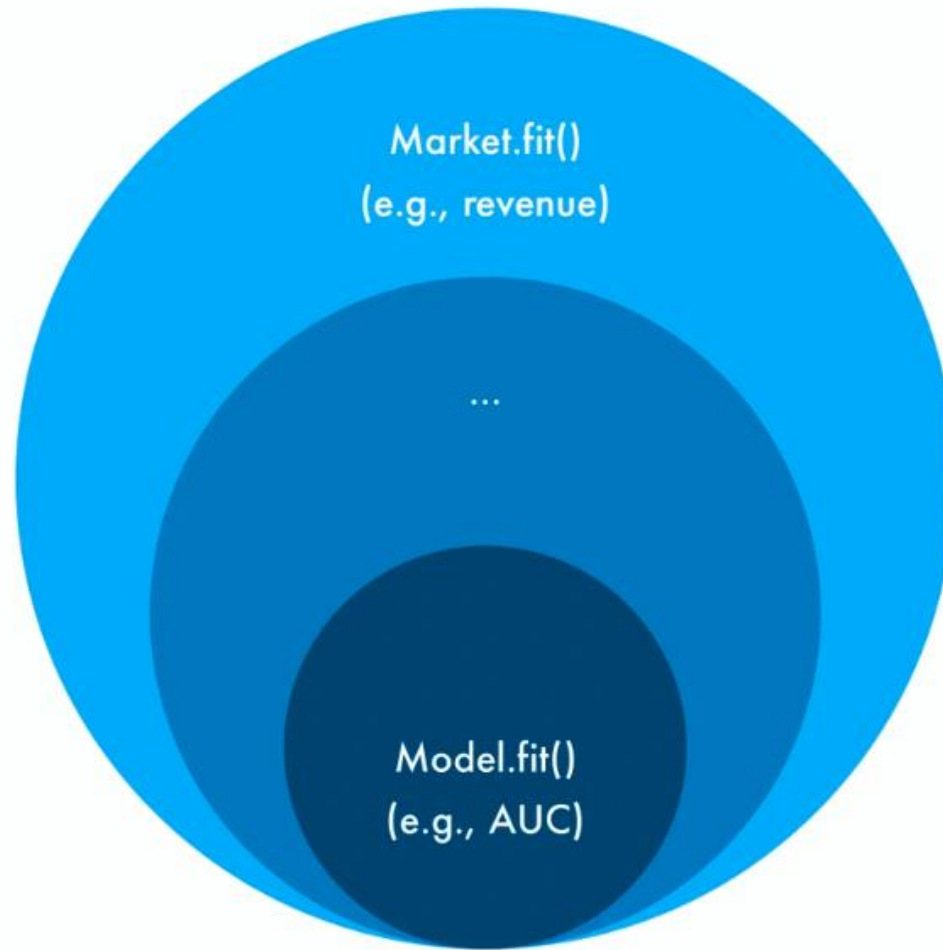
Case study: Tesla

Telsa is an AI-first company, with the ultimate goals of buildings the world's best AV ...

However, they began with a well-designed EV, equipped with the necessary array of sensor for autonomy. Over time, it can cruise control, automatically park, automatically break, and so on.



PMF is known to be the key driver of products' success and failure



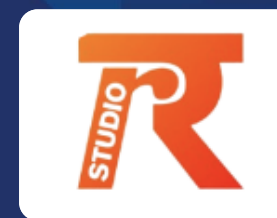


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Section 2: Product-market fit

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What is product-market fit?

A market is a set of related customer needs.

A product is a specific offering intended to meet a certain set of customer needs

Product-market fit is a measure of how well a product satisfies the market.



Why products fail

Products we build will exist in the solution space.

Problems, on the other hand, define the market.

Iteratively testing and improving the products' hypotheses is preferred over prematurely jumping to the solution space.



Customers don't care about your solution. They care about their problems.

Dave McClure

• outelands

<https://leanproductplaybook.com/>

First thing first: Who is your customer

- There are many ways to define the customers; for instance, one can segment the users' market based on demographics, behavior, needs, ...
- Another approach is to define personas; this means making many as many assumptions in order to simplify the overall market.
- Note that, there can be difference between users and buyers.



Identify understand customer needs

A good discovery process aims to illuminate the problem space, despite customers being much more aware, articulate and opinionated in the solution space.

Frameworks such as $GAP = Importance - Satisfaction$ can help to decide how to prioritize the value proposition to the customers; the bigger the gap the more underserved the need/opportunity.

Customer Benefit	Typical Customer Comment
1. Help me prepare my tax return	"I don't really know much about taxes. I try to follow the instructions but they're confusing. I'm not sure which forms I should be filling out."
2. Check the accuracy of my tax return	"I'm not that great at math, so I know I'm probably making several mistakes when I'm adding and subtracting all those numbers on my tax forms."
3. Reduce my audit risk	"I'm worried about being audited but don't really know how risky my tax return is. It would be great to know if it would raise any yellow flags with the IRS so I could fix those parts."
4. Reduce the time it takes me to enter my tax information	"I spend lots of time each year entering data from all the tax forms I receive from my employer, bank, and brokerages."
5. Reduce the time it takes me to file my taxes	"I normally print my tax return and then go to the post office, wait in line, and mail it so I can get delivery confirmation. It would be great if I could avoid that hassle."
6. Maximize my tax deductions	"I don't know about all the deductions that I'm eligible to take. I'm probably leaving some money on the table."

Can ML predict the $P(\text{audit})$?

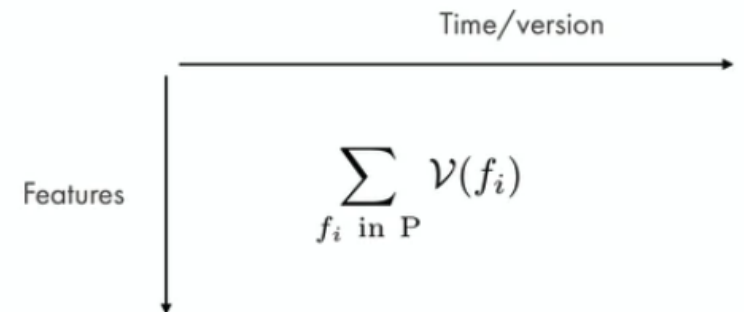
Can ML automate some tasks?



What are your MVPs (i.e., minimum viable product)?

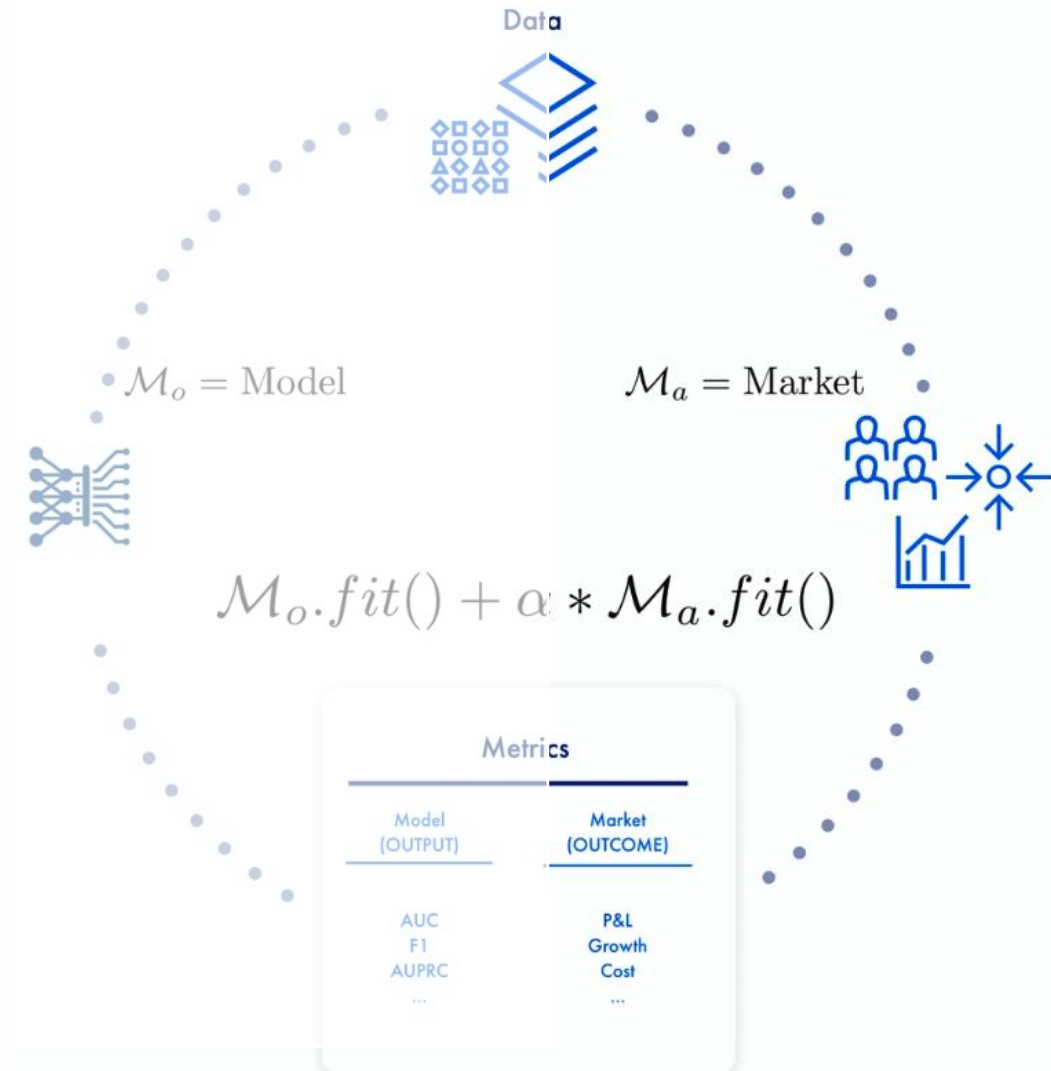
Given the list of value in mind, we can design a product value proposition template, which can summarize various MVP hypotheses, and competitors in one place – based on this, we can prioritize

	Competitor A		My Product	
	Now	In 1 Year	Now	In 1 Year
Must-Haves				
Must-have 1	Y	Y	Y	Y
Must-have 2	Y	Y	Y	Y
Performance Benefits				
Performance benefit 1	High	High	Medium	High
Performance benefit 2	Medium	High	Low	Low
Performance benefit 3	Low	Medium	High	High
Delighters				
Delighter 1	Y	Y		
Delighter 2			Y	Y
Delighter 3		Y		
Delighter 4				Y



A quick recap: So far, we have not build, rather have been exploring the problem space and hypothesizing potential solutions

- Formed hypotheses about your target customers and their underserved needs.
- Articulated the value proposition you plan to pursue so that your product is better and different.
- Identified the top feature ideas you believe will address those needs and broken them down to smaller chunks.
- Prioritized those features chunks based on ROI.
- Selected a set of those features for your MVP candidate/hypothesis, which you hypothesize customers will find valuable



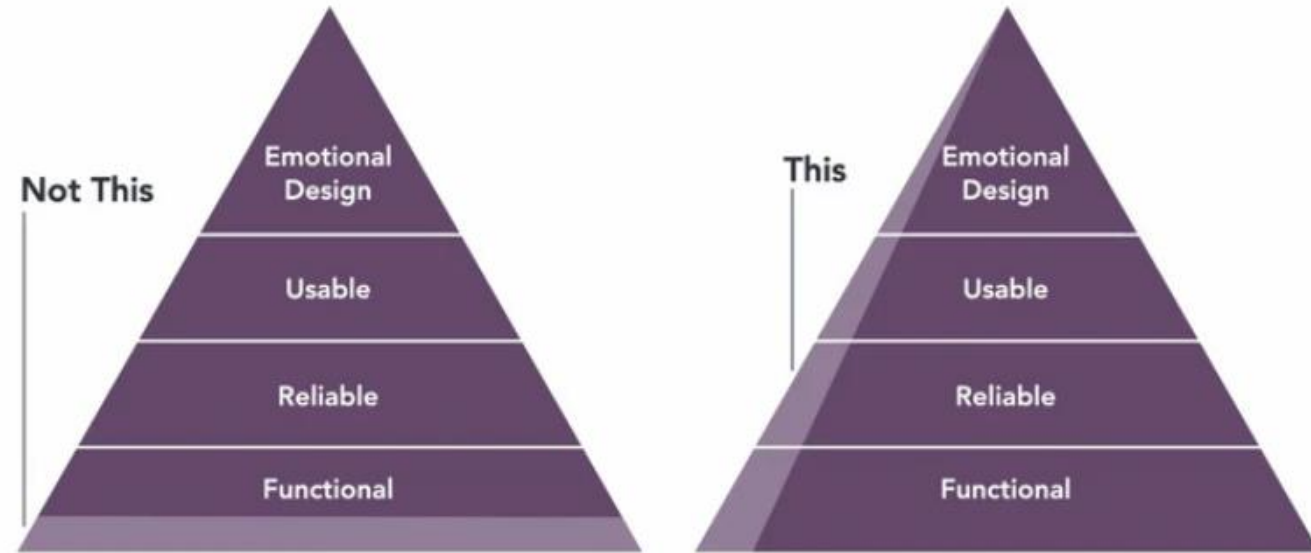
Let's hear from Steve Jobs



It's now time to build – starting with the MVP prototype

Many debate what does and does not qualify as MVP; while the terms minimum gets a lot of attention, it being a viable product is also important.

An MVP should not be viewed as minimally functional parodic and rather a combination of all key attributes.





The process of testing MVP hypotheses

- Test can be both qualitative and quantitative
- e Wizard of Oz and Concierge MVP allow you to actually test live product or service; instead of the final version, you are using manual workarounds – common for AI.

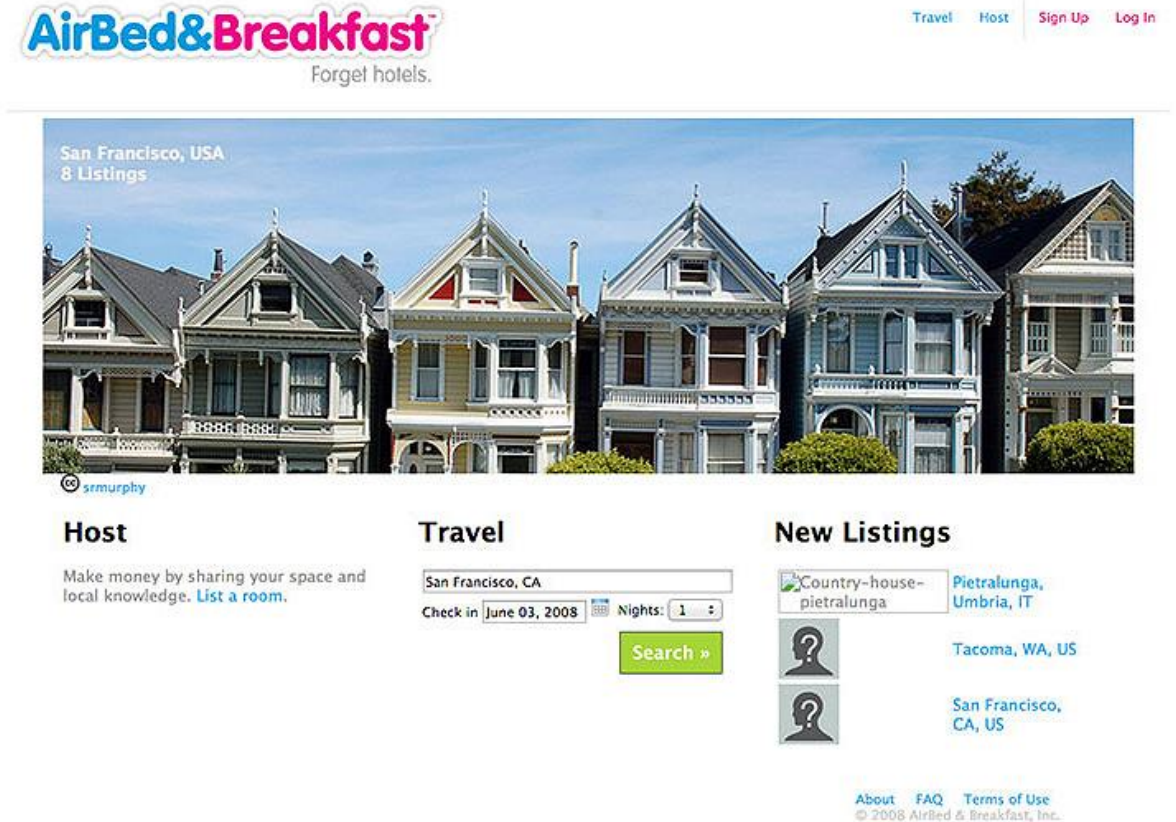
		Qualitative Tests	Quantitative Tests
Product Tests	Marketing Tests	Marketing materials	Landing page/Smoke test Explainer video Ad campaign Marketing A/B tests Crowdfunding
	Product Tests	Wireframes Mockups Interactive prototype Wizard of Oz & Concierge Live product	Fake door/404 page Product analytics & A/B tests

MVP Example: AirBnB

One of the hypotheses behind AirBnb idea was that property listings with professional photos would get more bookings.

Hence, they manually recruited the hosts, and photographers.

Next, they manually uploaded them into the website as if the real product was live.



MVP Explainer Video: DropBox



<https://www.yansmedia.com/blog/dropboxs-mvp-explainer-video-case-study>

AI-Washing Could Be on the Rise

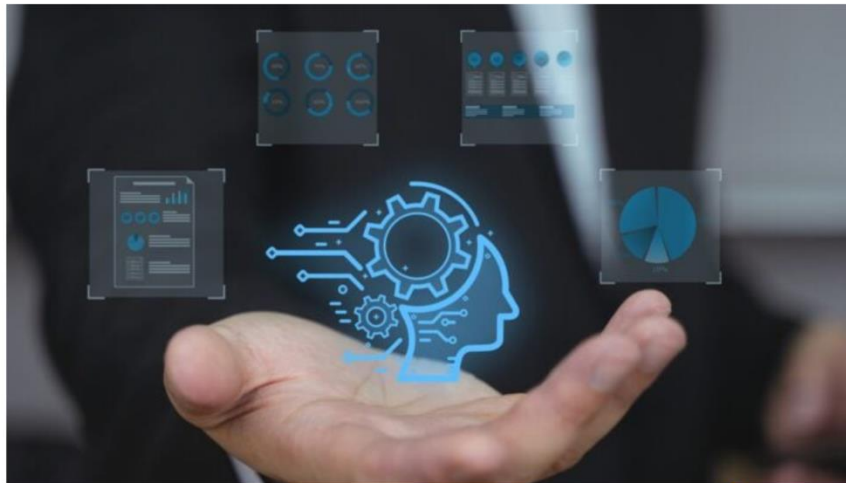
AI Chatbot Turns Out to Be 700 Engineers in India

The company has declared for bankruptcy and owes millions to Amazon and Microsoft in cloud computing costs.



Written by
Nicole Mousicos

Published on
June 6, 2025



London-based startup Builder.ai reportedly hired 700 engineers **to pose as, and do the work of, its AI chatbot** Natasha. The company initially promised customers the ability to produce an app entirely generated by AI.

Most Recent

Study: Entry-Level Employment Has Dropped By 13% Due to AI

Nicole Mousicos - 6 hours ago

Robomart's Autonomous Delivery Robot Challenges Delivery Platforms

Nicole Mousicos - 10 hours ago

Coinbase CEO Fires AI-Averse Engineers After Just One Week

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Musk's xAI Sues OpenAI and Apple for Being Anticompetitive

Adam Rowe - 1 day ago

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Adam Rowe - 2 days ago

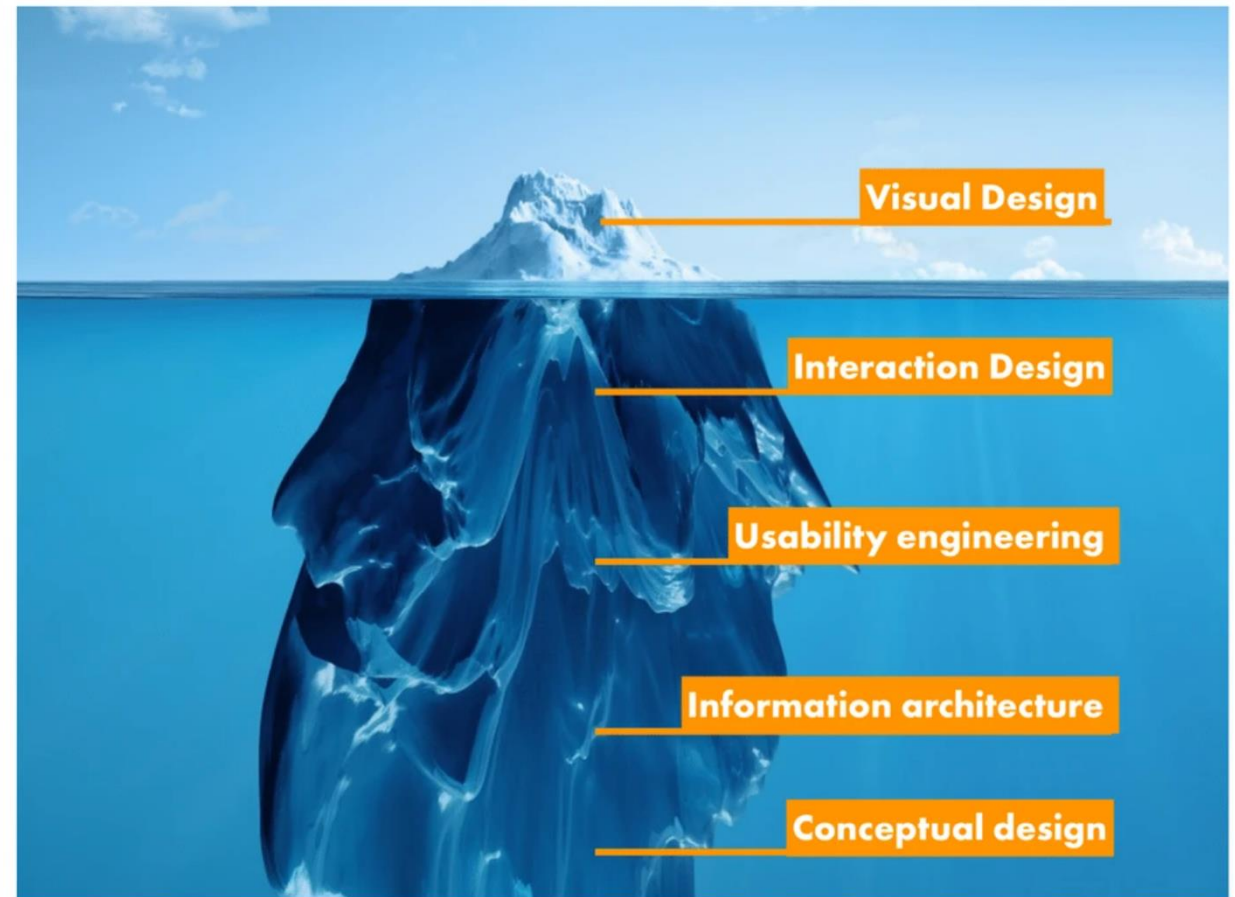
<https://tech.co/news/ai-startup-chatbot-revealed-as-human-engineers>

Now, the product can be tested with actual customer – build, measure, and learn loop (+ UX design)

It is crucial as you conduct your user test to differentiate between feedback on usability vs product-market fit.

Feedback on usability has to do with how easy it is for customers to understand and use your product, whereas feedback on product-market fit has to do with how valuable they find your product.

Considering pivoting when necessary!



Andrew Ng's Advice on Product Testing



- Ask 3 friends or team members to look at the product and let you know what they think (this might take ~ 0.5 days).
- Ask 10 friends or team members to take a look (~ 2 days).
- Send it to 100 trusted/volunteer alpha testers (~ 1 week?).
- Send it to 1,000 users to get qualitative or quantitative feedback (~ 2 weeks?).
- Incorporate it into an existing product to get feedback (1 to 2 months?).
- Roll it out to a large user base of an existing product and do rigorous A/B testing.



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Section 3: Build, Measure, and Learn

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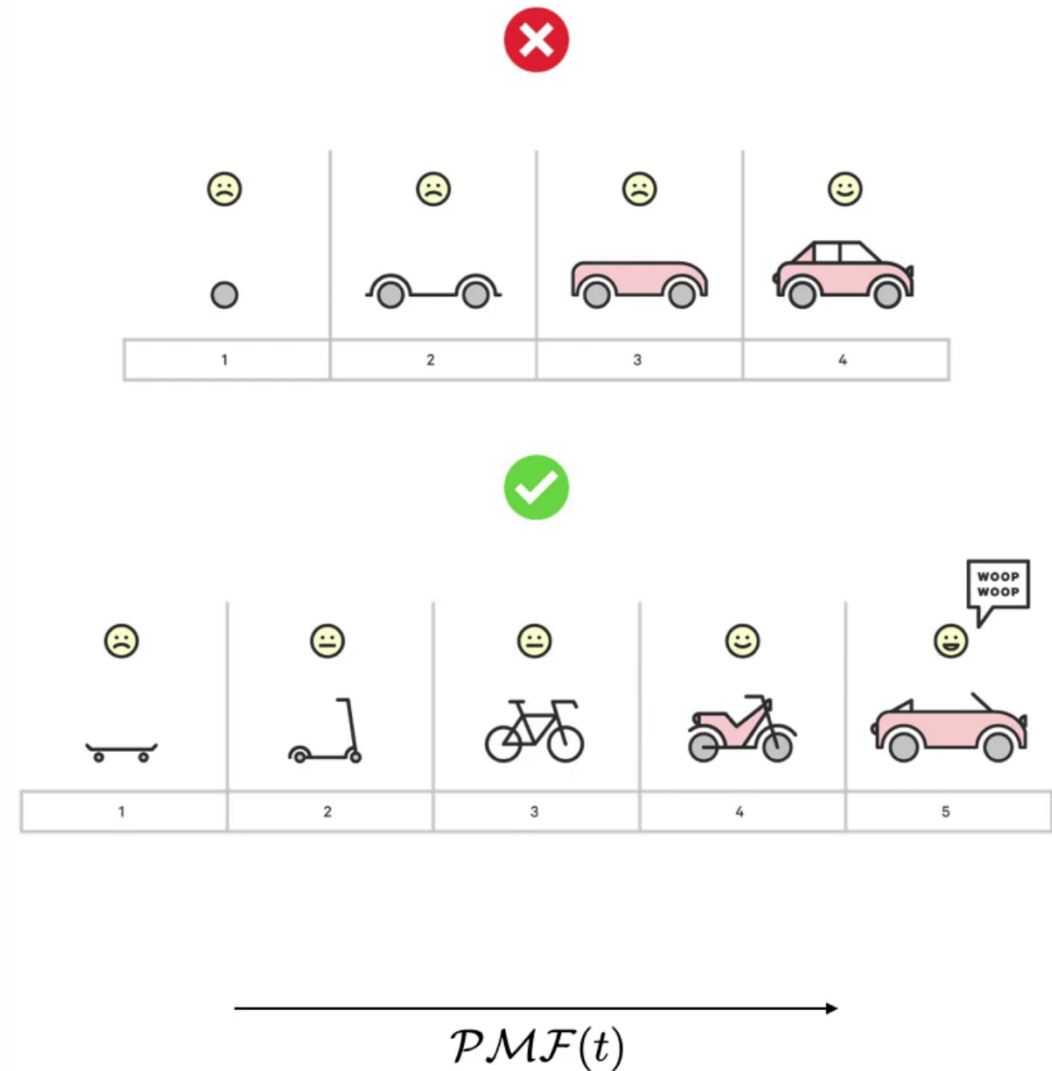


Perturb/intervene and measure

Given the so-called market in most application domains of AI is not computationally irreducible, the only way to know its reaction to our idea/product to to “perturb/intervene and measure”.

“If you are not embarrassed by the first version of your product, you’ve launched too late”

Reid Hoffman



Agile encourages early and continuous delivery of working software

“We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

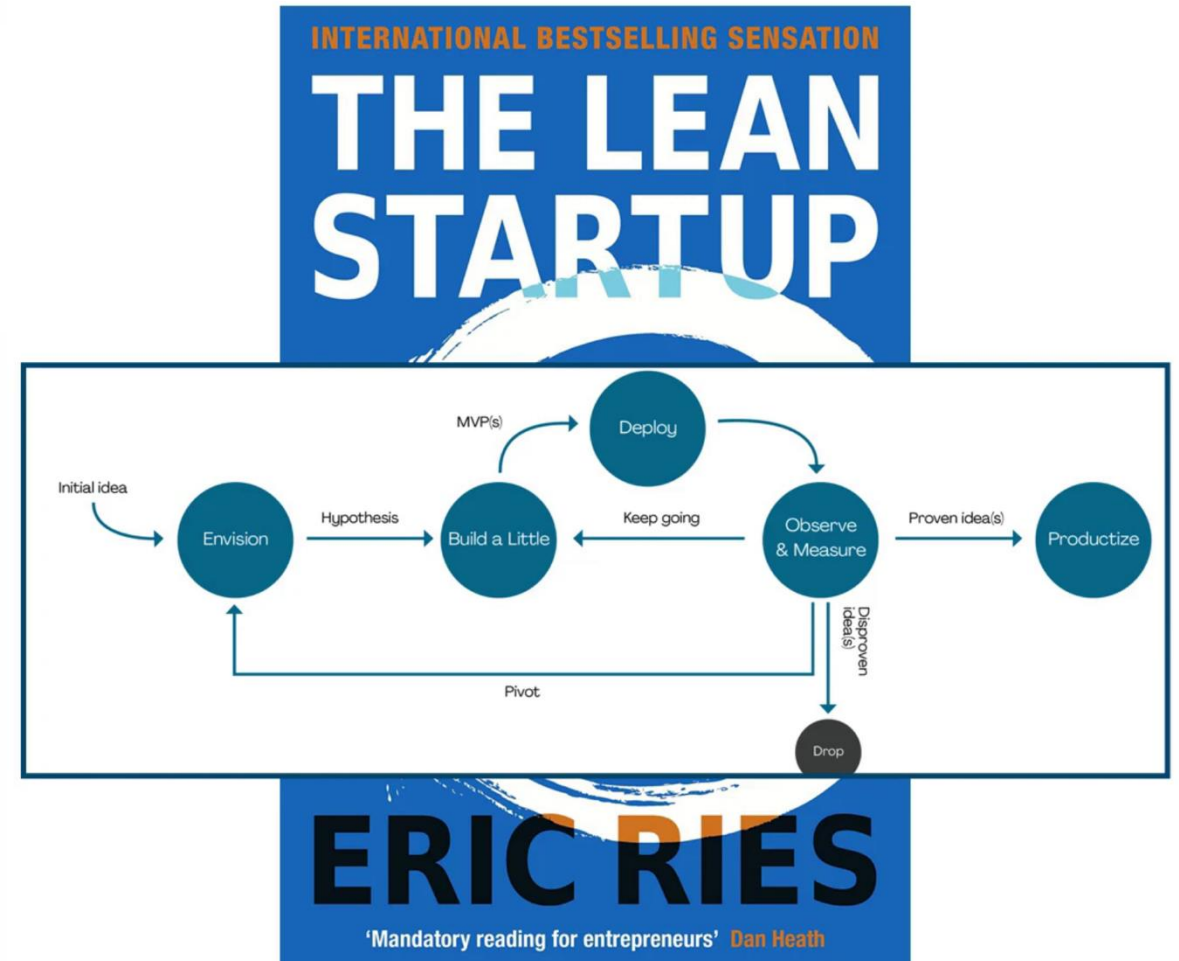
- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan”

<https://agilemanifesto.org/>

Experiment in order to discover/build the right thing that customers are willing to pay for

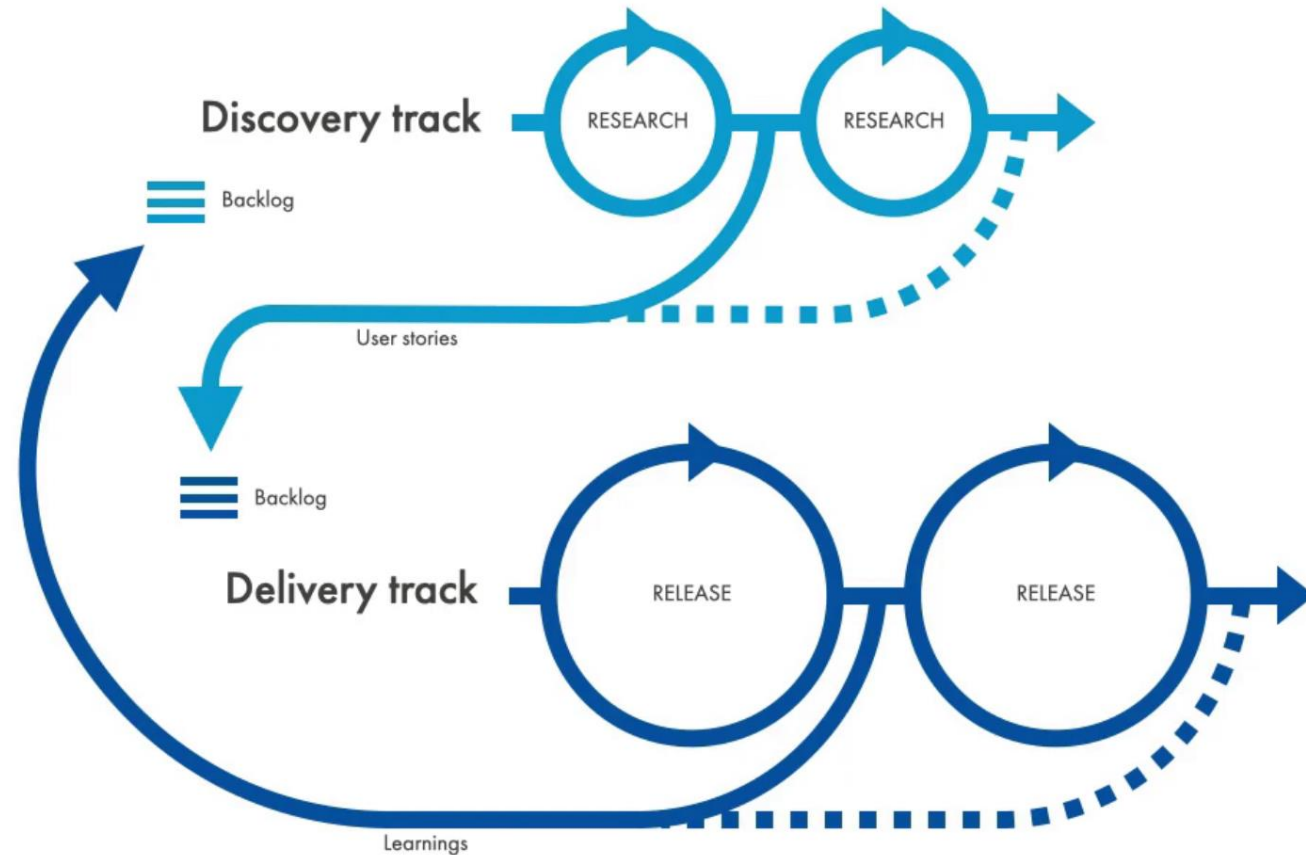
A project is not complete after you ship a version – as only then you receive user feedback from real-world interactions, which enables you to refine the goals for the next iteration.

As a consequence - you should strive to execute very short iterations focused on learning what the user wants.



Successful products benefit from vision, strategy, objectives, and discovery and delivery

- Discovery work focuses on fast learning and validation with the same agile and lean principles in mind.
- Discovery is a necessary part of product development; its progress should be visible to the whole team.
- If we're doing discovery right, we substantially change and kill lots of ideas.
- Keep measuring and learning even after you ship.





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Go-To-Market Strategy

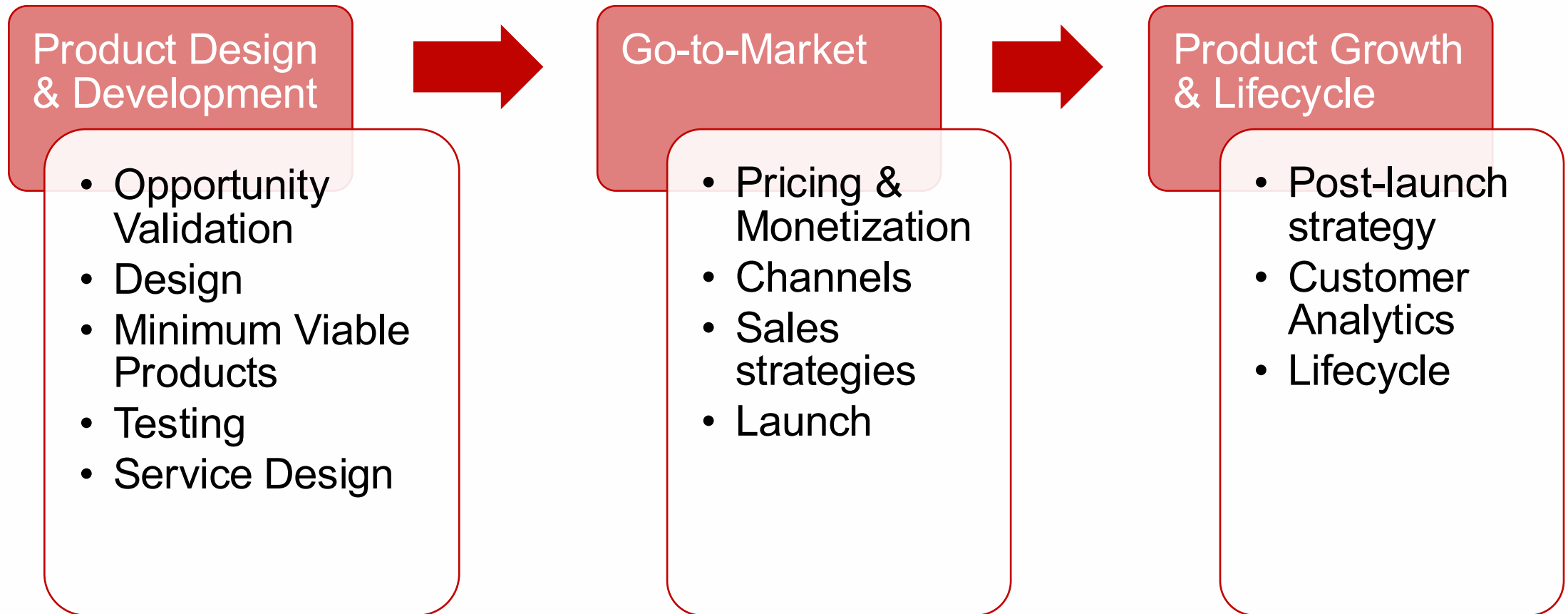
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Roadmap

We are chronologically and systematically following new product management process





Go-to-Market Strategy (1/3)

Your *action plan* for reaching your target customers:

Go-to-Market Strategy



marketing
channels



build awareness
& generate demand



sales
channels



close sales



distribution
channels



deliver the
solution to
customers

“freemium” pricing

<https://slack.com/pricing>

Make teamwork more productive

		BEST VALUE			
Free		Pro		Business+	
A simpler way to chat and collaborate		Drive productivity in one place		Scale with AI-powered work	
		50% OFF FOR 3 MONTHS*		50% OFF FOR 3 MONTHS*	
\$0 USD		\$8.75 \$4.38 USD		\$18 \$9 USD	
free forever		per user / month, when paying monthly ⓘ		per user / month, when paying monthly ⓘ	
GET STARTED		GET STARTED		GET STARTED	
✓ 90 days of message history ⓘ ✓ Up to 10 apps ⓘ ✓ 1:1 meetings ⓘ ✓ 1:1 external messages ⓘ		✓ Unlimited message history ⓘ ✓ Unlimited app integrations ⓘ ✓ Group meetings ⓘ ✓ Group external messages ⓘ		✓ Unlimited message history ⓘ ✓ Unlimited app integrations ⓘ ✓ Group meetings ⓘ ✓ Group external messages ⓘ	
 Basic AI		 Basic AI		 Advanced AI	
				 Contact sales for pricing	
				CONTACT SALES	
				 Enterprise-Grade AI	

Go-to-Market Strategy (2/3)

Based on the answers to these questions:

- Which customers will you target?
- Which of their needs will you address?
- What kind of relationship you will have with the customers?
- How will they buy from you?

Go-to-Market Strategy (3/3)

Your *initial* go-to-market strategy may be different from what it will be at scale.

- Early adopters vs. mainstream customers
- Market awareness
- Budgetary constraints

Tesla Roadster

Step 1: Start with Luxury, Not the Mass Market

The conventional approach to disruptive technology is to start cheap, gain market adoption, then scale up.

Tesla flipped that strategy upside down.

Instead of launching with an affordable EV, Tesla targeted the high end of the market with a \$100,000+ Roadster, attracting early adopters and technology enthusiasts.



Target Customer and Building Awareness

B2B vs. B2C

- Buyer vs end user
- Job title or position
- Influencers

Ways to make your customers aware that your solution is available?

- Website (or landing page)
- Personal networks
- Referrals



Step 1: Log into Dropbox

[Click here](#) to sign in and go directly to the **Refer a friend** tab on your account settings page.



Step 2: Make referrals

Send a referral to anyone by email. Or copy your custom invite link and paste it anywhere you like.



Step 3: Get free space

Once a referral creates a new account, you both get extra storage: 500 MB for Basic users and 1 GB for Plus.

Dropbox referral program FAQs:

How do I send a referral?



How do I earn free storage?



How much free space can I earn?



Dropbox used a **simple landing page and referral program** (“get 500MB free storage for inviting friends”).

Targeting tech-savvy professionals at first, leveraging awareness through personal networks and referrals rather than ads.

Outbound Marketing

The goal of **marketing** is to generate leads.

Traditional outbound channels include:

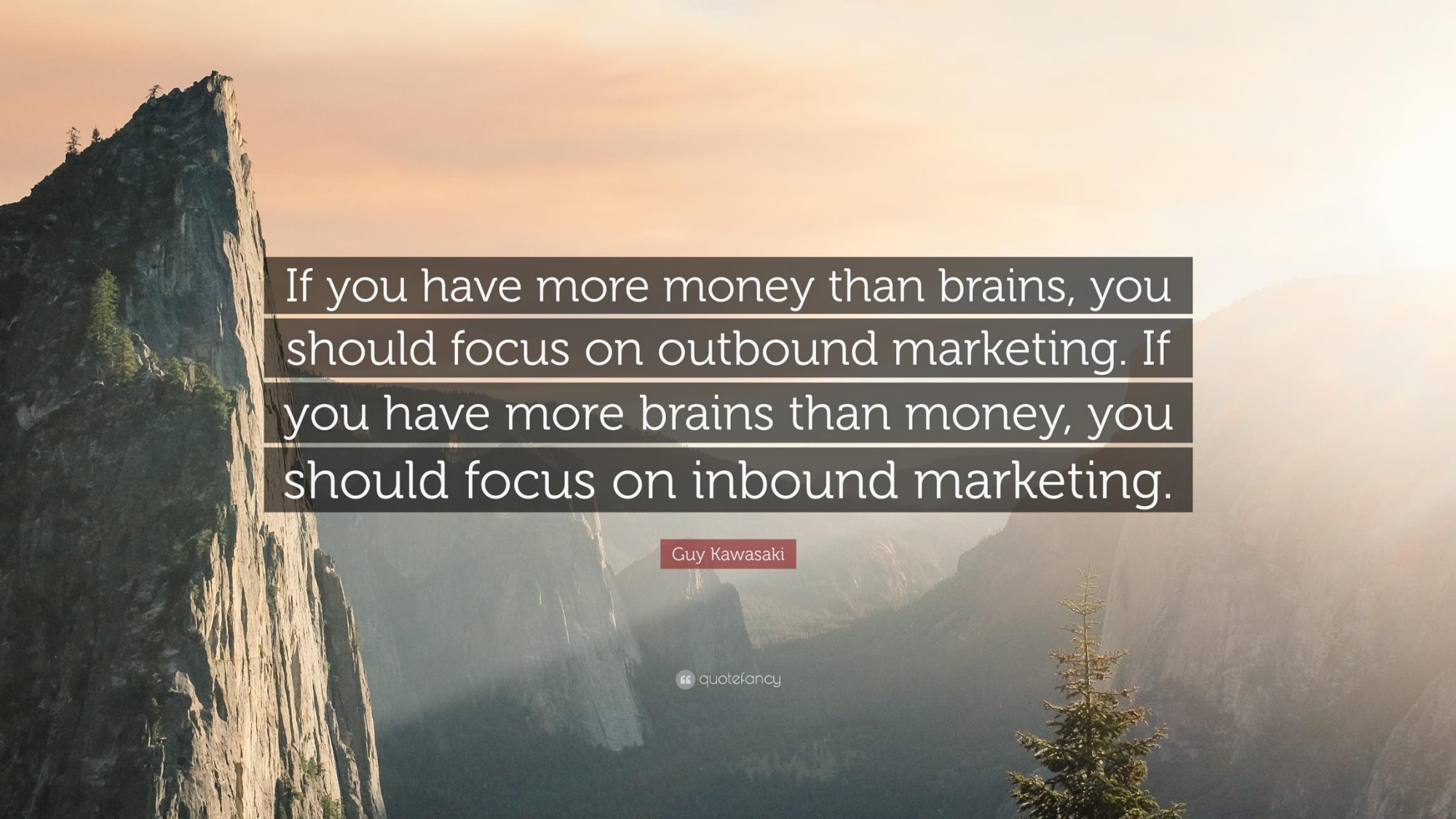
- Advertising
- Public relations
- Direct mail, emails, phone calls
- In-person sales calls

Inbound Marketing

Ways to help customers who are **actively seeking a solution** to find you:

- Word of mouth
- SEO
- Content creation: company/personal blogs; newsletters, white papers, etc.
- Social media
- Events

Inbound Marketing

A scenic view of a mountain peak, likely El Capitan in Yosemite National Park, with a quote overlay. The image shows a steep, rocky cliff face on the left, with a few trees growing on its lower slopes. The background is a hazy, sunlit valley with more mountains in the distance. The sky is a soft, warm orange and yellow, suggesting sunrise or sunset. A quote is overlaid in the center of the image, enclosed in a dark rectangular box with white text. The quote is attributed to Guy Kawasaki and is about inbound and outbound marketing.

If you have more money than brains, you should focus on outbound marketing. If you have more brains than money, you should focus on inbound marketing.

Guy Kawasaki

quote fancy

Physical/Online Sales Channels

- Direct sales
- Independent sales representations
- Distributors
- Retailers
- Etc.
- E-commerce
- Online stores
- App stores
- Market places
- Etc.

Channel Selection

- Depend on the customer relationship: one-time, on-going relationship, need training/support, etc.
- Customer expectations/preferences: stand-alone product or part of a bundled solution? Direct buy from you or through an aggregator?
- Channel capabilities

Metrics

Ways of monitoring the effectiveness of your go-to-market strategy:

- Revenue per dollar of sales expense
- Conversation rate
- Length of sales cycle
- Customer Acquisition Cost (CAC)

Vanity Metrics

- “likes”
- Followers
- Page views
- Email open rate
- Free downloads